

Donation & Reuse Platform

For Clothes and Household Items

Professional Project Specification & Software Requirements Document

Project Type	Web-Based Social Impact Platform
Primary Users	Donors, NGOs, Beneficiaries, Administrators
Frontend	React.js, HTML5, CSS3, JavaScript, Tailwind CSS
Backend	Node.js, Express.js
Database	PostgreSQL
Deployment	Vercel + Render
Phase 1	Web-only; no native mobile app or payment handling

1. Project Overview

The Donation & Reuse Platform is a web-based social impact solution that connects donors with verified NGOs and beneficiaries. The platform provides a centralized way to donate usable clothes and household items while improving verification, collection coordination, status tracking, and transparency.

2. Context and Problem Statement

Context: The platform connects donors with verified NGOs and individuals who can benefit from reusable items.

Problem Statement: Existing donation processes often suffer from a lack of trusted/verified channels, logistics and doorstep-collection challenges, and manual tracking of donation requests and outcomes.

3. Project Objectives

- **Centralized platform:** Provide a single digital platform for listing, requesting, collecting, and tracking donations.
- **Donor-NGO connection:** Connect donors with verified organizations and beneficiaries.
- **Doorstep collection:** Enable donors to schedule convenient collection requests.
- **Transparency:** Maintain clear donation statuses and donation history.
- **Scalability:** Design the platform so it can expand across multiple cities.

4. Scope of the Product

In Scope	Out of Scope – Phase 1
Donor registration and profiles	Native Android/iOS applications

NGO registration and verification	Payment processing / cash handling
Beneficiary management	Advanced logistics fleet management
Item listing and donation requests	Full international/multi-country deployment
Doorstep collection scheduling	Complex reward marketplace
Donation status tracking	AI matching in initial release
Admin dashboard and dispute handling	

5. User Roles

Role	Main Responsibilities
Donor	Register, create profile, list items, schedule collection, track donations, view history.
NGO	Maintain verified profile, review requests, accept donations, coordinate collection, update statuses.
Beneficiary	Maintain profile, submit/receive eligible requests, view relevant donation status.
Admin	Verify NGOs, monitor users and collections, manage disputes, maintain platform oversight.

6. Functional Requirements

Donor Features

- User registration and secure login
- Profile creation and management
- Add/list clothes and household items
- Upload item information and condition
- Select pickup address and preferred time
- Schedule doorstep collection
- Track donation status
- View donation history

NGO / Beneficiary Features

- Create and manage organization/beneficiary profile
- NGO verification workflow
- View available donation requests
- Accept or reject suitable requests
- Coordinate collections
- Update donation lifecycle/status
- Maintain receiving/beneficiary records

Admin Features

- Admin authentication and role-based access

- Verify and approve/reject NGOs
- Monitor users, donations, and collections
- Review suspicious or disputed activities
- Manage disputes and platform reports
- View platform-level statistics and KPIs

7. Non-Functional Requirements

Category	Requirement
Performance	Donation and status updates should normally be processed/displayed within approximately 3 seconds under expected load.
Usability	Interfaces should be simple, responsive, accessible, and easy for first-time users.
Security	Secure authentication, password hashing, authorization, validation, and protected APIs.
Scalability	Architecture should support expansion from one city to multiple cities.
Availability	Core platform services should be designed for reliable access and graceful error handling.
Maintainability	Use modular frontend/backend architecture and documented REST APIs.

8. Technical Architecture

Frontend: React.js with Tailwind CSS, HTML5, CSS3 and modern JavaScript. The frontend provides responsive dashboards, forms, item listings, status tracking, and administrative interfaces.

Backend: Node.js with Express.js. The backend exposes REST APIs for authentication, users, NGOs, items, donations, collections, notifications, and administration.

Database: PostgreSQL stores relational application data with appropriate primary keys, foreign keys, indexes, constraints, and timestamps.

External APIs: Maps/location services can support address selection, geolocation, distance calculation, and collection planning. Notification services may be added for email or alerts.

Deployment: Frontend can be deployed on Vercel and backend on Render, with PostgreSQL hosted through a suitable managed database service.

9. Suggested Database Design

Entity	Important Data
users	id, name, email, password_hash, role, phone, status, created_at
donors	id, user_id, address_id, preferences
ngos	id, user_id, organization_name, registration_details, verification_status
beneficiaries	id, user_id, needs, address_id, verification_status
addresses	id, user_id, address_line, city, state, postal_code, latitude, longitude
items	id, donor_id, category, title, description, condition, quantity, image_url, status
donations	id, item_id, donor_id, ngo_id, beneficiary_id, status, accepted_at, completed_at
collection_requests	id, donation_id, address_id, preferred_date, preferred_time, status
notifications	id, user_id, type, message, is_read, created_at
disputes	id, donation_id, raised_by, reason, status, resolution
admin_actions	id, admin_id, action_type, target_type, target_id, notes, created_at

10. High-Level User Flow

1. Donor registers and logs in.
2. Donor creates/updates profile and address.
3. Donor lists an item with category, condition, quantity, and details.
4. Donor submits a donation request and selects a collection preference.
5. Eligible verified NGOs/beneficiaries review the request.
6. NGO accepts the donation request.
7. Collection request is scheduled and tracked.
8. Item is collected/received.
9. NGO updates the donation status.
10. Donation is marked completed and added to the donor's history.

11. REST API Structure

Module	Example Endpoints
Authentication	POST /api/auth/register, POST /api/auth/login
Users	GET /api/users/me, PUT /api/users/me
NGOs	POST /api/ngos, GET /api/ngos, PATCH /api/ngos/:id/verify
Items	POST /api/items, GET /api/items, PUT /api/items/:id
Donations	POST /api/donations, GET /api/donations/:id, PATCH /api/donations/:id/status
Collections	POST /api/collections, GET /api/collections, PATCH /api/collections/:id/status
Notifications	GET /api/notifications, PATCH /api/notifications/:id/read
Admin	GET /api/admin/dashboard, GET /api/admin/disputes, PATCH /api/admin/disputes/:id

12. Security Design

- JWT-based authentication for protected sessions/APIs.
- Password hashing using a secure password-hashing algorithm.
- Role-Based Access Control (RBAC) for Donor, NGO, Beneficiary, and Admin permissions.
- Request validation and sanitization on server-side APIs.
- Parameterized database queries/ORM protections against SQL injection.
- Rate limiting and secure error responses for public APIs.
- HTTPS in production and secure handling of environment variables/secrets.
- Audit logging for important administrative actions.

13. Admin Dashboard Requirements

- Dashboard summary: users, verified NGOs, active donations, completed donations, pending collections.
- NGO verification queue with approve/reject actions.
- Donation and collection monitoring.
- User and role management.
- Dispute management and resolution tracking.
- Reports and KPI visualization.
- Audit history for important administrative actions.

14. Project KPIs

KPI	Purpose
Registered Users	Measure platform adoption.
Verified NGOs	Measure the strength of the verified partner network.
Total Donations	Measure overall donation activity.
Completed Collections	Measure operational success.
Repeat Donation Rate	Measure donor retention and engagement.
Collection Success Rate	Measure how effectively scheduled collections are completed.
User Satisfaction	Measure user experience and trust.
Items Reused/Redirected	Measure environmental and social impact.

15. Project Assumptions

- Donors provide accurate item and contact information.
- NGOs provide valid registration/verification information.
- Users have access to a working internet connection.
- Collection availability depends on participating NGOs and operational capacity.
- Phase 1 does not process monetary donations or payments.

16. Project Constraints

- Initial release is web-only.
- No native mobile application in Phase 1.
- No cash/payment handling in Phase 1.
- Logistics are coordinated through the platform but may depend on NGO operations.
- Expansion to multiple cities requires additional operational and verification processes.

17. Deliverables

- Functional responsive web application
- Donor onboarding and dashboard
- NGO/beneficiary onboarding and dashboard
- Admin dashboard
- Donation and item management module
- Collection scheduling and tracking module
- PostgreSQL database
- REST API backend
- Authentication and authorization system
- Deployment-ready production build
- Technical documentation

18. Future Enhancements

- **Native Mobile App:** Android/iOS applications for easier access and real-time notifications.
- **AI-Based Matching:** Match donated items with NGOs/beneficiaries based on item category, location, demand, and urgency.
- **AI Item Classification:** Use image-based classification to identify item categories and condition.
- **Smart Logistics:** Optimize collection routes and pickup scheduling.
- **Gamification:** Badges, milestones, leaderboards, and impact scores for donors.
- **Impact Analytics:** Show estimated waste diverted, items reused, and community impact.
- **Multi-City Expansion:** Support city-wise NGO networks, collection zones, and localized administration.

19. Expected Social & Environmental Impact

The platform aims to reduce usable-item waste by making donation easier and more organized. It can improve trust through NGO verification and transparent status tracking, increase community participation, and help usable clothes and household items reach people and organizations that need them.

20. Recommended Development Roadmap

Phase	Work
Phase 1	Requirements, UI/UX design, database schema, project setup.
Phase 2	Authentication, roles, user profiles, NGO onboarding.
Phase 3	Item listing, donation requests, image handling.
Phase 4	Collection scheduling, location APIs, status tracking.
Phase 5	Admin dashboard, NGO verification, disputes, notifications.
Phase 6	Testing, security hardening, performance optimization.
Phase 7	Deployment, documentation, monitoring, final demonstration.

Conclusion

The Donation & Reuse Platform provides a practical full-stack solution for improving the donation lifecycle—from listing an item and finding a suitable verified recipient to doorstep collection and completion tracking. The proposed architecture is modular, secure, and suitable for future expansion with mobile applications, AI-powered matching, smart logistics, and gamification.